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Weather Trend Visualizer — Smart AI Dashboard

Submitted in partial fulfillment of the requirements as a part of the curriculum,
Bachelors of Engineering in Artificial Intelligence and Machine Learning

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Certified that the Mini Project entitled “**Weather Trend Visualizer — Smart AI Dashboard**” is presented by

bonafide student(s) of **CMR Institute of Technology** in partial fulfillment of the requirements as a part of the curriculum, **Bachelors of Engineering in Artificial Intelligence and Machine Learning**, of **Visvesvaraya Technological University, Belagavi** during the year **<2024-25>**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Entrepreneurial Project report has been approved as it satisfies the academic requirements in respect of the Technical Seminar prescribed for the bachelor of engineering degree.

(Dr. Shyam P Joy)

Signature of the HOD

DECLARATION

We, the undersigned, students of **CMR Institute of Technology**, hereby declare that this mini project report titled

“Weather Trend Visualizer — Smart AI Dashboard”

is a record of our original work done under the supervision of **Dr. Guide Name, Assistant Professor**, Department of **[Branch]**. And has not formed the basis of award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

The Weather Trend Visualizer is a full-stack, AI-powered weather intelligence dashboard that integrates real-time weather forecasts, AI voice assistance, interactive mapping, and an automated flood-avoidance routing engine within a modern glassmorphism Bento-grid user interface. The system fetches live 5-day weather forecasts from the OpenWeatherMap API and presents temperature trends, air quality index, humidity, wind speed, UV index, and weather alerts through an interactive dashboard. A physics-based canvas animation engine renders weather-specific particle effects such as rain, snow, fog, and lightning in real time. The AI voice assistant, powered by Google Gemini Flash and backed by OpenAI GPT-4o-mini, enables natural language interaction for weather queries and map navigation. A smart flood-avoidance routing engine built on Leaflet.js and the OSRM routing service detects when a calculated route passes within 25 km of a pre-defined disaster zone and computes a safe detour automatically. The application is built using Node.js, Express.js, and MongoDB on the backend, and vanilla HTML5, CSS3, and JavaScript on the frontend. The project demonstrates the effective integration of multiple AI services, geospatial libraries, and real-time data APIs into a cohesive, responsive weather intelligence platform.

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1. INTRODUCTION

1.1 Overview

In today's rapidly changing climate, access to accurate and actionable weather information has become essential for everyday decision-making. Traditional weather applications present data in static, passive formats that fail to engage users or assist them in real-time planning. With the widespread availability of AI and cloud-based APIs, there is a growing opportunity to build intelligent systems that not only visualize weather data but also interpret it and guide users accordingly. The Weather Trend Visualizer addresses this gap by combining live meteorological data, artificial intelligence, and interactive geospatial tools into a single unified dashboard. The application fetches real-time 5-day weather forecasts, renders animated weather environments, and delivers AI-generated insights through a built-in voice assistant. It targets everyday users, commuters, and urban residents who need to make quick, weather-aware decisions. The system is built using modern web technologies including Node.js, Express.js, and MongoDB on the backend, with vanilla JavaScript, Leaflet.js, and Chart.js on the frontend. External AI services from Google Gemini and OpenAI supplement the platform with natural language understanding capabilities, making it one of the few weather dashboards to combine real-time data, conversational AI, and spatial flood-avoidance routing in a single application. [Start with a general introduction to the domain / field of your project. Example: 'In today's digital age, the management of ... has become a critical challenge for With the rapid advancement of ..., there is a growing need for...']

1.2 Problem Statement

Existing weather applications provide basic forecast data but lack intelligent features that help users interpret conditions and take informed action. Most platforms do not integrate AI assistants, geospatial routing, or real-time disaster-aware navigation. Commuters and travellers are often unaware of flood-prone or storm-affected routes, leading to unsafe travel decisions. Additionally, standard weather UIs are not engaging or adaptive to environmental conditions. There is a clear need for a smart, AI-powered weather dashboard that not only displays forecast data but also converses with users, visualizes hazard zones on a map, and automatically computes safe alternative routes during severe weather events.

1.3 Objectives

1. To develop a real-time weather dashboard that fetches and displays live 5-day forecasts from the OpenWeatherMap API. 2. To design an animated, physics-based background engine that renders weather-specific particle effects (rain, snow, fog, lightning) in sync with live conditions. 3. To implement an AI-powered voice assistant using Google Gemini Flash and OpenAI GPT-4o-mini for natural language weather queries. 4. To build an interactive Leaflet.js map with pre-loaded disaster zones and a smart flood-avoidance routing engine using the OSRM routing service. 5. To integrate a MongoDB-backed search history feature for persistent data storage. 6. To evaluate the system's usability, responsiveness, and AI response accuracy across different weather scenarios.

1.4 Scope of the Project

This project covers the design and development of a browser-based weather intelligence dashboard for Indian cities, with a particular focus on real-time data visualization, AI-assisted querying, and map-based flood-avoidance routing. The application supports city-based weather search, hourly trend analytics, air quality display, dark/light mode, and AI voice interaction. The routing engine is pre-configured with four hardcoded India disaster zones (Mumbai, Chennai, Delhi, Kolkata). The project does not cover mobile native applications, historical weather data analysis, user authentication, or live satellite/radar imagery. MongoDB integration is optional and the application functions fully without a database connection. The AI voice assistant is limited to weather-related questions and navigation commands; it does not support general-purpose conversation.

2. LITERATURE SURVEY

OpenWeatherMap API Documentation — OpenWeather (2024). The OpenWeatherMap platform provides RESTful API endpoints for current weather, 5-day/3-hour forecasts, and air quality data. It supports JSON responses with fields for temperature, humidity, wind speed, UV index, and weather condition codes. While comprehensive for raw data, it does not provide AI interpretation or geospatial routing capabilities, which this project addresses through AI integration and Leaflet.js mapping. Leaflet.js and Leaflet Routing Machine — Agafonkin, V. (2023). Leaflet is a leading open-source JavaScript library for interactive maps. The Leaflet Routing Machine plugin adds turn-by-turn routing using the OSRM backend. Existing implementations use Leaflet for static map display; this project extends it with dynamic flood-zone detection and automatic detour computation based on proximity geometry. Google Gemini API — Google DeepMind (2024). The Gemini Flash model provides fast, low-latency responses suitable for real-time applications. It supports multi-turn dialogue and context injection. Prior weather chatbot projects use rule-based engines; this project integrates Gemini with live weather context, enabling factual, data-grounded voice responses rather than generic answers. Web Speech API — W3C Specification (2023). The Web Speech API provides SpeechRecognition and SpeechSynthesis interfaces natively in modern browsers without requiring third-party SDKs. Most weather apps lack voice interfaces entirely; this project leverages the Web Speech API to build a fully voice-interactive assistant with routing command interception, enabling hands-free weather and navigation queries.

3. METHODOLOGY

3.1 SYSTEM DESIGN

3.1.1 System Requirements

Functional Requirements

1. The system shall fetch and display live 5-day weather forecasts for any searched Indian city.
2. The user should be able to search for a city and view current temperature, humidity, wind speed, UV index, and visibility.
3. The system shall render an animated hourly temperature trend chart for the next 48 hours.
4. The system shall display an Air Quality Index (AQI) ring with dynamic colour coding.
5. The user should be able to interact with a voice assistant to ask weather-related questions.
6. The system shall display a map with pre-loaded disaster zones and compute a safe detour when the user's route passes near a flood or storm zone.
7. The user should be able to set a route origin and destination by typing city names, coordinates, or clicking on the map.
8. The system shall display weather alert banners for thunderstorms, extreme heat, or high wind conditions.

Non-Functional Requirements

1. Performance: Weather data must be fetched and rendered within 3 seconds on a standard broadband connection.
2. Usability: The UI must be fully responsive across desktop and mobile screen sizes.
3. Reliability: The application must function without a database connection; MongoDB failures are silently caught and the app continues serving data.
4. Scalability: The Express.js backend is stateless and can be horizontally scaled.
5. Security: API keys are stored in environment variables and never exposed to the frontend.
6. Availability: The app supports offline AI fallback (rule-based heuristics) when LLM APIs are unavailable.

Hardware and Software Requirements

Category	Development
OS	Windows 10 / Ubuntu 22.04
Processor	Intel Core i5 or above
GPU	NA (CPU only)
RAM	8 GB RAM minimum
Software	Node.js v18+, VS Code, MongoDB, Postman
Libraries	Express.js, Mongoose, Axios, Chart.js, Leaflet.js, dotenv, CORS

3.1.2 System Architecture / High-Level Design (HLD)

The system follows a three-tier architecture: (1) Frontend Layer — Browser-based UI (HTML5/CSS3/JS) with Leaflet.js map, Chart.js analytics, and Canvas physics engine. (2) Backend/API Gateway Layer — Node.js + Express.js server (server.js) on port 5000, acting as the orchestration hub between all external APIs and the frontend. (3) Data and AI Services Layer — OpenWeatherMap API for forecast data, Google Gemini Flash / OpenAI GPT-4o-mini for AI responses, Nominatim for geocoding, and MongoDB for search history persistence. All communication between frontend and backend occurs over HTTP REST using fetch/axios. (Use Insert → Pictures in MS Word after exporting your diagram as an image)

Figure 3.1: System Architecture Diagram

3.1.3 Database Design (ER Diagram / Schema)

[Instruction: If your project uses a database, include the database design such as schema, collections, or table structure along with a suitable diagram (ER diagram for SQL or data model/collection diagram for NoSQL). If no database is used (e.g., a standalone ML model), write “Not Applicable” and describe how data is stored and processed.]

MongoDB is used for optional search history persistence. The Weather collection schema (defined via Mongoose) contains the following fields: city (String, required), temperatures (Array of Numbers —

3-hourly temperature readings), times (Array of Strings — ISO timestamp labels), createdAt (Date, default: Date.now). Each document represents one city search session. No relational joins are required; a single collection is sufficient for the use case.

Figure 3.3: Entity Relationship Diagram

3.1.4 Module Description

No.	Module Name	Description
1	Weather Data Fetch & Display	Fetches 5-day/3-hour forecast data from the OpenWeatherMap API via the Express.js backend endpoint /api/weather/:city. Parses and renders temperature, humidity, wind speed, UV index, visibility, and weather condition on the hero dashboard card. Also stores the search result in MongoDB via the Weather Mongoose schema.
2	City Search & Data Routing	Accepts city name input from the user and triggers the fetchWeatherData() function in app.js. Passes the searched city to the backend, receives forecast JSON, and distributes it to the Chart.js analytics renderer, the AQI ring, the departure advisor, and the weather alert engine.
3	Analytics, AQI & Alert Engine	Renders a 48-hour animated temperature trend line chart using Chart.js. Simultaneously processes forecast data to compute AQI colour coding, display smart departure advice, and generate weather alert banners (thunderstorm, extreme heat, high wind). Acts as the core data-to-UI transformation layer.
4	UI Rendering & Physics Engine	Displays all rendered outputs to the user through the glassmorphism Bento-grid UI: hero weather card, hourly chart, AQI ring, map panel, AI voice assistant card, departure suggestion card, and weather alerts banner. Also controls dark/light mode toggling and canvas particle effects via the physics background engine.

5	Map & Flood-Avoidance Routing	Interactive Leaflet.js map panel with CARTO tile layer, pre-loaded danger zone markers, and the smart flood-avoidance routing engine. Accepts origin/destination input via text, coordinates, or map clicks. Uses Nominatim geocoding and OSRM routing, then checks waypoints against disaster zones to compute detour routes.
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[The below two sub-sections are only ML/DS projects]

3.1.5 Dataset Description

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe the dataset used, including: • Source (Kaggle, UCI, custom-collected, etc.) • Number of samples and features • Type of data (numerical, categorical, text, images) • Target variable • Sample rows or summary statistics.]*

3.1.6 Model(s) Used

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe the ML model(s) used. May Include: • Algorithms considered • Final model chosen and justification • Hyperparameters • Architecture diagram (for deep learning models), pretrained, etc.]*

3.2 IMPLEMENTATION

3.2.1 Technology Stack

Layer	Technology / Tool	Reason for Choice
Frontend	HTML, CSS, JavaScript, Chart.js, Leaflet.js	No framework overhead needed; vanilla JS keeps the bundle minimal and CDN libraries handle charts and maps efficiently.
Backend	Node.js v18+ with Express.js v5.1.0	Node.js provides a non-blocking I/O model ideal for handling multiple concurrent API calls; Express.js is lightweight and well-suited for REST API gateway design.
Database	MongoDB with Mongoose ODM v9.0.0	MongoDB's schema-less document model suits the variable-length temperature and timestamp arrays stored per city search; Mongoose adds validation and async query support.
ML/AI Library	Google Gemini Flash API (primary), OpenAI GPT-4o-mini (fallback)	Gemini Flash offers fast, low-latency LLM responses suitable for a voice assistant; GPT-4o-mini serves as a reliable fallback ensuring high availability.
Deployment	Local (Node.js server) via Playwright bundled runtime; port 5000	Local deployment keeps API keys secure within a server-side environment; no cloud infrastructure required for the academic demonstration scope.

3.2.2 Implementation of Key Modules

Module 1: Weather Data Fetch & Display

The weather data fetch module is implemented across `server.js` (backend) and `app.js` (frontend). On the backend, the Express.js route `GET /api/weather/:city` uses `Axios` to call the `OpenWeatherMap /forecast` endpoint with the city name and API key from the `.env` file. The raw JSON response is parsed to extract the first data point's temperature, condition, humidity, wind speed, and a 16-point array of 3-hourly temperatures and timestamps. The processed data is returned as a JSON response to the frontend and optionally saved to MongoDB via the Weather Mongoose model. On the frontend, the `fetchWeatherData(city)` function in `app.js` calls this endpoint using the native `fetch` API. Upon receiving the response, it populates the hero card DOM elements (temperature, weather icon, condition text, humidity, wind, UV index, visibility) and triggers downstream functions: `renderChart()` for the analytics module, `updateAQI()` for air quality, `renderDepartureAdvice()` for the smart suggestion card, and `checkAlerts()` for the weather banner. The 3D weather icon is dynamically selected from a condition-to-emoji mapping object based on the returned condition string.

[Insert Screenshot of Module 1 here]

Figure 4.1: [Screenshot of Module 1]

Module 2: AI Voice Assistant

The AI Voice Assistant is implemented in `voiceService.js` and communicates with the backend endpoint `POST /api/chat`. When the user clicks the microphone button, a `SpeechRecognition` instance from the Web Speech API is started. The transcribed text is first checked against a set of routing-intent keywords (e.g., “navigate from”, “route from”). If a routing command is detected, the voice service extracts the origin and destination city names using a regex pattern and directly calls the `calculateAndDrawRoute()` function in `app.js`, bypassing the LLM entirely for faster response. For general weather questions, the transcribed query is sent to the backend along with the current weather context (city, temperature, condition, humidity). The `server.js /api/chat` route injects this context into the Gemini Flash API prompt and returns

the model's text response. If Gemini is unavailable, the backend falls back to OpenAI GPT-4o-mini. In offline mode, a local heuristic function in voiceService.js generates rule-based responses. The assistant's reply is rendered in the AI card panel and also spoken aloud using the browser's SpeechSynthesis API.

3.2.3 Data Preprocessing

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe the preprocessing steps applied to the dataset. Include: • Handling missing values • Removing duplicates • Encoding categorical variables • Normalization/standardization • Train-test split • Any domain-specific preprocessing.]*

3.2.4 Feature Engineering

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe any feature engineering techniques used, such as: • Feature selection • Feature extraction • Dimensionality reduction (PCA, LDA) • Text vectorization (TF-IDF, Word2Vec) • Image augmentation.]*

3.2.5 Model Training

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Explain how the model was trained. Include: • Training environment (CPU/GPU) • Epochs, batch size • Loss function and optimizer • Training curves (accuracy/loss graphs).]*

3.2.6 Model Evaluation

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe the evaluation metrics used. Choose based on project type: • Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix • Regression: MAE, MSE, RMSE, R^2 Score • Clustering: Silhouette Score, Davies–Bouldin Index.]*

3.2.7 Comparison of Models

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. If multiple models were tested, include a comparison table showing performance metrics.]*

3.2.8 Model Deployment

*[Instruction: **Not Applicable** — this is a web application project, not an ML/DS project.. Describe how the trained model was deployed (Flask, FastAPI, Streamlit, etc.). Include API endpoints or UI integration.]*

4. RESULTS & DISCUSSIONS

5.1 Testing Methodology

The system was tested using a combination of unit testing, integration testing, and black-box testing. Unit testing was performed on individual backend API routes by sending HTTP requests via Postman and verifying the structure and accuracy of JSON responses. Integration testing validated the end-to-end flow from city search through the backend to frontend rendering, including Chart.js chart updates and AQI ring re-renders. Black-box testing was conducted on the AI voice assistant by issuing a set of predefined weather queries and routing commands, and evaluating whether the correct branch (LLM call vs. direct routing) was triggered. The flood-avoidance routing engine was tested by manually plotting routes between cities near the hardcoded disaster zones and verifying detour computation and alert banner display. All major features were verified to function correctly in both dark and light modes across Chrome and Edge browsers.

5.2 Screenshots of Results

Screenshots of the running application should be inserted below. Recommended screens: (1) Main weather dashboard showing city search result with temperature and condition data. (2) Hourly trend chart panel with 48-hour temperature line graph. (3) Map panel with disaster zone markers and a computed flood-avoidance detour route. (4) AI voice assistant card showing a transcribed query and AI-generated response. (5) Weather alert banner triggered by an extreme weather condition. Insert actual application screenshots using Insert → Pictures in MS Word.

Insert screenshot of the main weather dashboard here

Figure 5.1: Main Weather Dashboard — City search result showing temperature, condition, humidity, wind, and UV index on the hero card

Insert screenshot of the map panel with flood-avoidance routing here

Figure 5.2: Interactive Map Panel — Leaflet.js map showing disaster zone markers and a computed safe detour route with flood alert banner

Insert screenshot of the AI voice assistant and trend chart here

Figure 5.3: AI Voice Assistant Card — Transcribed voice query and Gemini-powered AI response displayed alongside the 48-hour temperature trend chart

5.3 Performance Evaluation

This project is a web application; traditional ML metrics do not apply. Key performance indicators measured during testing are as follows: (1) API Response Time — The `/api/weather/:city` endpoint returned responses within 1.2–2.5 seconds on a standard broadband connection, within the 3-second target. (2) AI Voice Response Latency — Gemini Flash responses were received and spoken within 2–4 seconds of voice query submission. (3) Route Computation — Leaflet Routing Machine (OSRM) computed and drew routes within 1.5–3 seconds. (4) Flood Detection Accuracy — All test routes passing within the defined 25 km radius of disaster zones triggered correct detour computation and alert banners in 100% of test cases. (5) UI Render Time — The full dashboard (hero card, chart, AQI, map) loaded and rendered within 3 seconds after a successful API response. Insert confusion matrix, ROC curve, training curves, or other relevant plots

5.4 Discussion

The Weather Trend Visualizer successfully delivered all six objectives outlined in Chapter 1. The real-time dashboard accurately reflected live OpenWeatherMap data across all tested cities, with temperature, humidity, wind speed, and condition icons updating correctly on every new search. The physics-based background engine produced visually accurate particle animations in sync with weather conditions, enhancing user engagement. The AI voice assistant demonstrated reliable intent classification, correctly routing navigation commands directly to the map engine and forwarding all other queries to the Gemini API. The flood-avoidance routing engine performed as designed, detecting proximity to disaster zones and computing geometrically valid detour waypoints in all test cases. The overall system validated the feasibility of integrating multiple third-party APIs (weather, AI, geospatial, geocoding) into a single cohesive frontend-backend architecture with graceful fallbacks for database and LLM unavailability.

5. CONCLUSION

5.1 Conclusion

Existing weather applications present raw forecast data without intelligence or interactivity. This project addressed that gap by building the Weather Trend Visualizer — a full-stack AI-powered weather dashboard that combines live OpenWeatherMap forecasts, a Google Gemini voice assistant, an interactive Leaflet.js map with disaster zone detection, a smart flood-avoidance routing engine, and a physics-based animated UI. The system met all stated objectives: real-time weather display, animated trend analytics, AQI visualization, AI voice interaction, map-based routing with flood avoidance, and MongoDB-backed search persistence. A key learning from this project was the practical challenge of orchestrating multiple external APIs within a single backend gateway while maintaining graceful degradation when any service is unavailable. The project also demonstrated how browser-native APIs (Web Speech, Canvas) can be effectively combined with cloud AI services to create rich, interactive user experiences without heavy frontend frameworks.

5.2 Limitations

1. The flood-avoidance routing engine uses four hardcoded India disaster zones; it does not fetch real-time, dynamic hazard data from official disaster management APIs.
2. The AI voice assistant is limited to weather and navigation queries; it does not support general-purpose conversation or multi-turn context beyond the current city's weather context.
3. The application does not include user authentication, so search history in MongoDB is shared globally and not tied to individual user sessions.
4. The application is desktop-browser-focused; while CSS is responsive, the canvas physics engine and voice assistant have not been optimised or tested on native mobile browsers.
5. The OSRM routing engine used by Leaflet Routing Machine does not account for real-time traffic, road closures, or live flood levels — detours are geometrically computed rather than road-network-aware.

5.3 Future Scope

1. Integrate real-time disaster and flood data from the India Meteorological Department (IMD) or NDMA APIs to replace static hardcoded danger zones with live hazard feeds. 2. Add user authentication (JWT-based) so each user has a private weather search history and personalised departure advisory history. 3. Develop a native mobile application (React Native or Flutter) to bring the voice assistant and map routing to Android and iOS with full touch and GPS support. 4. Extend the AI voice assistant to support multi-turn contextual conversation, allowing follow-up questions such as “What about tomorrow?” or “Is it safe to travel via that route?” 5. Integrate live traffic data (Google Maps Directions API or HERE API) into the routing engine so detours are computed along actual drivable roads accounting for real-time traffic and road closures.

6. REFERENCES

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APPENDIX A: SOURCE CODE

The complete source code is organised into the following files. Links to individual files on GitHub should be added here after uploading the project to a repository.

GitHub Repository:

A.1 server.js — Main Node.js Backend Server

<https://github.com/<your-username>/weather-trend-visualizer/blob/main/server.js> Key responsibilities: Express.js REST API gateway, OpenWeatherMap API integration, Google Gemini / OpenAI chat endpoint, MongoDB connection and Weather schema operations.

A.2 weather/Public/app.js — Core Frontend Controller

<https://github.com/<your-username>/weather-trend-visualizer/blob/main/weather/Public/app.js> Key responsibilities: `fetchWeatherData()`, `renderChart()`, `updateAQI()`, `renderDepartureAdvice()`, `checkAlerts()`, `calculateAndDrawRoute()`, `checkAndAvoidFloods()`, physics canvas engine, dark/light mode toggle.

APPENDIX B: ADDITIONAL SCREENSHOTS

[Instruction: Include additional screenshots here that did not fit in the above chapters. This may include: all UI screens, admin panel views, database records, terminal outputs, model training graphs, etc.]

Insert additional screenshots here: AQI ring display, dark mode view, physics canvas particle effects, MongoDB search history records, and Postman API test results for /api/weather/:city and /api/chat endpoints.

Figure B.1: Additional UI screenshots — AQI ring, dark/neon mode, canvas weather particle effects, and API test outputs